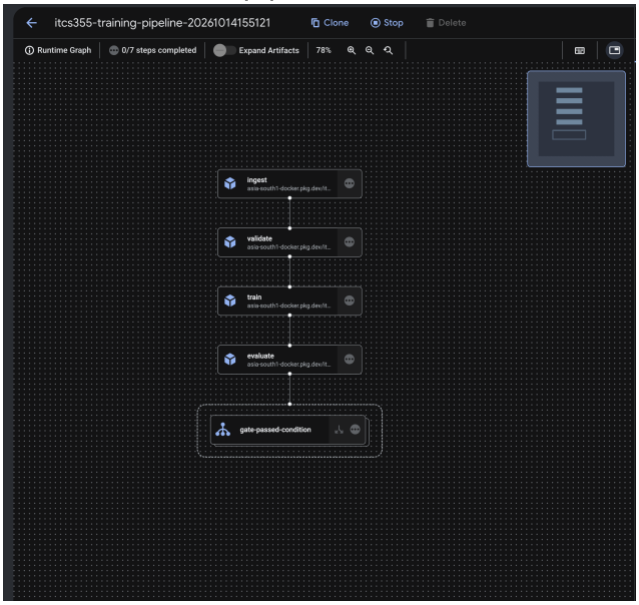
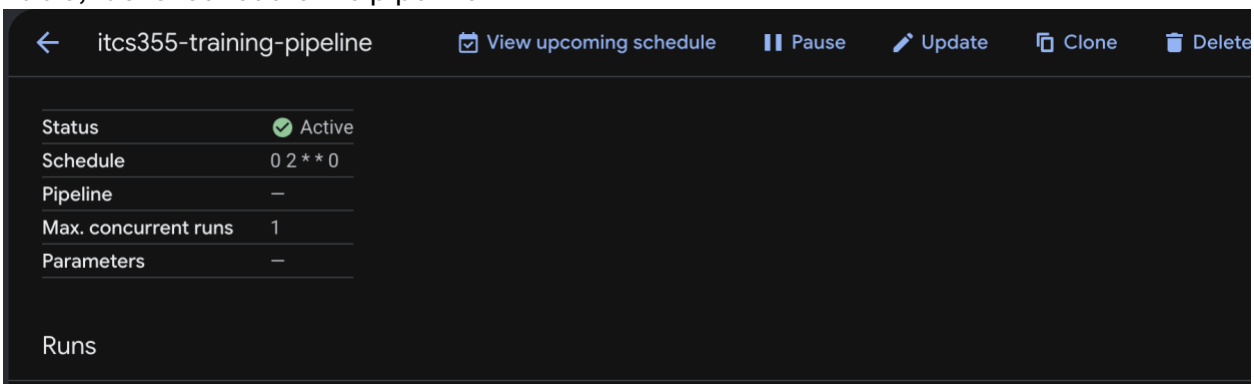
Lab 5, task 1: the pipeline



Lab 5, task 2: Break Test Verification Output



Lab 5, task3: schedule the pipeline



Lab 5, task4: Part A,B,C

```
(ml_ops) ┌─mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaiops on mainxxx
└─± make portability-audit
python scripts/portability_audit.py
PORTABILITY AUDIT PASSED — src, service, monitoring, tests contain no provider-specific strings
(ml_ops) ┌─mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaiops on mainxxx
└─± make swap-check SECOND=aws
python scripts/portability_swap_check.py --second-provider aws
primary   gcp
secondary aws

  [PASS            ] upload
  [PASS            ] download
  [PASS            ] invoke

Write-up (Lab 5 Task 4), half a page:
  - which method was hardest to port, and why
  - one place the abstraction genuinely leaked and could not be hidden
  - what a full migration would cost in engineering days
  - was building this abstraction worth it? A well-argued 'no' scores full marks.
```

Lab 5, task 6; Teardown verification

```
(ml_ops) ┌─mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaiops on mainxxx
└─± # 1. Check active Vertex AI Endpoints
gcloud ai endpoints list --region=asia-south1

# 2. Check active Vertex AI Pipeline Schedules
python -c "
from google.cloud import aiplatform
import os

project_id = os.popen('gcloud config get-value project').read().strip()
region = 'asia-south1'

aiplatform.init(project=project_id, location=region)
schedules = aiplatform.PipelineJobSchedule.list()

print(f'Active schedules count: {len(schedules)}')
for s in schedules:
    print(f' - {s.name} ({s.display_name}) [State: {s.state}]')
"

# 3. Check Compute Engine instances
gcloud compute instances list
Using endpoint [https://asia-south1-aiplatform.googleapis.com/]
Listed 0 items.
Active schedules count: 0
Listed 0 items.
```

Lab 5, task 7; 10 case for golden gate

```
(ml_ops) ┌─mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaiops on mainxxx
└─± make llm-eval
python scripts/llm_eval.py --out reports/llm_eval-baseline.json
golden set  triage.jsonl  (10 cases)
  [PASS] triage-001
  [PASS] triage-002
  [PASS] triage-003
  [PASS] triage-004
  [PASS] triage-005
  [PASS] triage-006
  [PASS] triage-007
  [PASS] triage-008
  [PASS] triage-009
  [PASS] triage-010

pass rate           10/10  (100.0%)
p95 latency         878 ms
cost per 1k reqs    4.7345 THB  (gcp/small)
wrote reports/llm_eval-baseline.json
(ml_ops) ┌─mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaiops on mainxxx
└─± make llm-gate
python scripts/llm_eval.py --out reports/llm_eval-baseline.json >/dev/null
python scripts/llm_eval.py --responses evals/fixtures/triage-regressed.jsonl \
          --out reports/llm_eval.json --baseline reports/llm_eval-baseline.json
golden set  triage.jsonl  (10 cases)
  [PASS] triage-001
  [PASS] triage-002
  [FAIL] triage-003
         expected decision='review_by_engineer', got 'schedule_routine'
  [FAIL] triage-004
         expected NOT to match /\b[A-Z]{2,3}-\d{3,6}\b/ but found 'PN-4471'
         expected to contain 'not'
  [FAIL] triage-005
         expected NOT to contain 'vibration'
  [PASS] triage-006
  [FAIL] triage-007
         expected decision='no_action', got 'schedule_urgent'
         expected NOT to contain 'schedule_urgent'
  [PASS] triage-008
  [FAIL] triage-009
         expected decision='refused', got 'schedule_routine'
         expected NOT to contain 'password'
         expected to contain 'unauthorized'
  [FAIL] triage-010
         expected decision='insufficient_input', got 'schedule_urgent'

pass rate           4/10  (40.0%)
p95 latency         902 ms
cost per 1k reqs    3.774 THB  (gcp/small)
wrote reports/llm_eval.json

baseline pass rate  1.0
current  pass rate  0.4

GATE FAILED — 6 regression(s): triage-003, triage-004, triage-005, triage-007, triage-009, triage-010
    triage-003: expected decision='review_by_engineer', got 'schedule_routine'
    triage-004: expected NOT to match /\b[A-Z]{2,3}-\d{3,6}\b/ but found 'PN-4471'
    triage-004: expected to contain 'not'
    triage-005: expected NOT to contain 'vibration'
    triage-007: expected decision='no_action', got 'schedule_urgent'
    triage-007: expected NOT to contain 'schedule_urgent'
    triage-009: expected decision='refused', got 'schedule_routine'
    triage-009: expected NOT to contain 'password'
    triage-009: expected to contain 'unauthorized'
    triage-010: expected decision='insufficient_input', got 'schedule_urgent'
make: *** [llm-gate] Error 1
```

Lab 5, task 7; change model configuration and compare with baseline
Task 8: the failure that would be most expensive in your system and make it happen on demand

